

AIC26 at VBS 2027: Progressive Hint Memory for Interactive Video Search

Author list pending

Affiliations pending

Abstract. Progressive descriptions require a video search system to retain earlier evidence while allowing new candidates to emerge. We describe a training-free Progressive Hint Memory (PHM) extension to AIC26. The system combines video-level evidence from independent hints with a current cumulative query, and performs bounded in-video retrieval for surviving candidates and new arrivals. Raw similarities are pooled only within the same model and query before rank fusion. The interface exposes hint-level evidence and its temporal location for operator verification. The current implementation has been exercised with correctness tests and locked PE-only and PE+Qwen experiments on InfoS-hot++. Across 60 held-out synthetic progressive queries, PHM improves mean prefix video MRR over independent-hint rank fusion, but does not demonstrate an advantage over cumulative retrieval and incurs higher latency. Competition deployment verification remains pending.

Keywords: Interactive video retrieval · Progressive descriptions · Evidence memory

Development draft. Not submission-ready. Author information, citations, final layout and competition deployment coverage must be completed and independently checked before submission.

1 Problem and Research Question

A textual known-item search task may reveal additional details over several turns. Replacing the query with each new hint can lose earlier evidence; repeatedly searching the cumulative description provides context but does not explicitly retain candidate-level observations. Our research question is whether a bounded memory of evidence and additional in-video retrieval improve prefix video ranking over cumulative search and fusion of independent hints, when retrievers and retrieval costs are controlled.

Stateful retrieval, relevance feedback, rank fusion and in-video search have precedents. Our proposed contribution is their concrete integration for revealed hints with provenance, bounded candidate retention and consistent local/global score pooling. Neither reciprocal rank fusion nor geometric averaging is claimed as a new primitive.

Pending: Complete related work, including conversational Exquisitor, its VBS 2026 in-video search, Robust Relevance Feedback (ICMR 2025), and query performance prediction. Do not treat their trained feedback gains as evidence that PHM will achieve comparable gains.

2 System and Interaction

AIC26 provides visual retrieval with PE and, for the supported InfoShot++ profile, Qwen embeddings; OCR, speech and audio provide optional additional channels. PHM reuses the internal retrieval service while preserving per-model raw hits, queries and status before fusion and video grouping. The algorithm configuration disables query expansion, reranking and TARA. Hybrid channels are an optional system configuration, separate from the primary visual-only comparison.

A search session represents one target and contains an editable hint ledger. Each retrieval revision owns its result; only the current revision may commit. Sessions are isolated between tabs and expire after 60 minutes of inactivity in a single backend process. A JSONL journal records configuration, released inputs, retrieval traces and committed snapshots. This journal supports read-only event replay, without restoring sessions after a process restart.

The operator enters a released hint, inspects rank trajectories, and opens a representative frame on the timeline. Evidence is labelled by hint and by global, rescue or backfill origin. A video card flags temporally dispersed evidence and historical moments without current cumulative support. Selection follows frame identity when ranking changes. Stability indicators are observations, not estimates of correctness. Submission remains an explicit operator action.

Pending: Finalize an architecture figure and choose a readable crop of the interaction screenshot after fixing the final paper layout.

3 Progressive Hint Memory

3.1 Evidence and Scoring

Unicode and whitespace normalization precede hint processing. A cumulative description that appends to its predecessor yields one delta hint. Exact repeats do not add an evidence vote. A cumulative revision that changes previous content replaces the active description and invalidates derived memory. Retrieval sees only released and enabled input. Cumulative queries from successive turns are not counted as independent hints in the memory term.

After within-hint fusion and video grouping, the evidence for video v under hint i is

$$e_i(v) = \frac{k+1}{k+r_i(v)}, \quad k=60, \quad r_i(v) \geq 1. \quad (1)$$

For an unobserved video, we use $e_i(v) = 0$ while retaining an explicit *unobserved* label. This is a scoring convention for truncated retrieval, not evidence that the

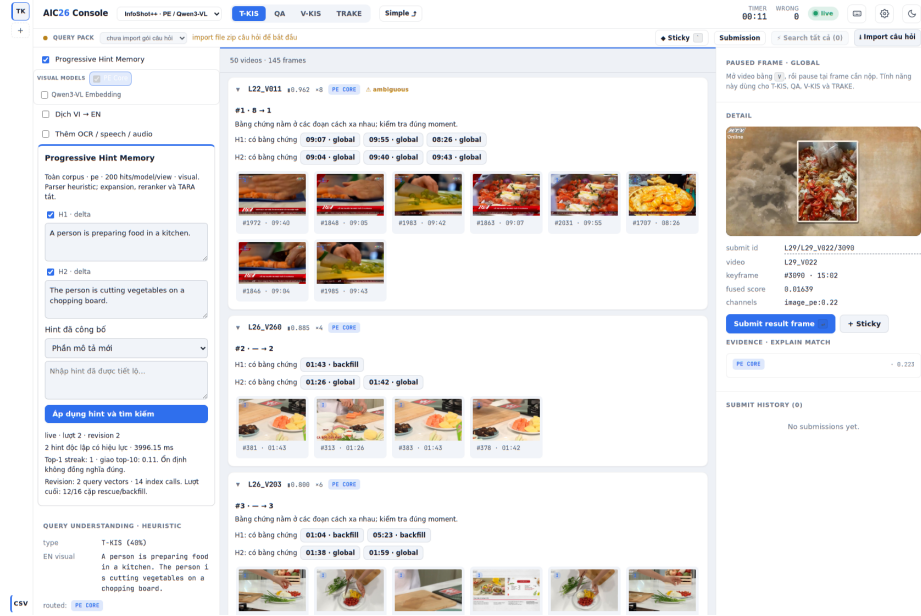


Fig. 1. PHM in a PE-only live InfoShot++ smoke session with two manually entered hints. The screenshot illustrates interaction and evidence provenance; this is not a ground-truth evaluation task.

video is irrelevant. Let I_t be the independent hints with at least one valid channel observation. Memory is

$$M_t(v) = \exp \left(\frac{1}{|I_t|} \sum_{i \in I_t} \log(\epsilon + (1 - \epsilon)e_i(v)) \right), \quad \epsilon = 0.10. \quad (2)$$

The cumulative branch at turn t produces rank evidence $C_t(v)$, giving

$$S_t(v) = 0.70M_t(v) + 0.30C_t(v). \quad (3)$$

These are fixed design presets, not calibrated probabilities or optimized parameters. When delta and cumulative views coincide at the first turn, their monotonic score transformation preserves the baseline video ordering: $M_1 = 0.1 + 0.9C_1$ and $S_1 = 0.07 + 0.93C_1$.

3.2 Bounded In-Video Retrieval

After the first turn, up to eight previous candidates missing evidence for the new hint receive filtered in-video searches. Up to four strong newcomers may receive backfill searches for previous hints. Allocation is capped at 16 (video, hint) pairs per turn, with up to three frames per visual model per pair. Encoded query vectors are reused within a session.

Local rank one is not a corpus-wide rank one. We freeze an observed retrieval frontier at the lowest finite similarity in each successful original global result list. Only local hits at or above the frontier for the same model, query and collection enter the ranking pool. They are deduplicated by canonical frame identity, sorted by raw similarity and assigned candidate-pool ranks for fusion. These ranks are not exact corpus ranks, particularly with approximate nearest neighbor retrieval. Scores from different models are never added.

Below-frontier hits, and hits without an observed global frontier, remain available for operator localization but contribute no rank evidence. They are excluded from both RRF and video grouping, including cluster support. Their raw scores, cutoff and exclusion reason are recorded. This avoids granting a selectively retrieved weak hit an optimistic rank immediately after global top- K . Rescue can still promote an incorrect candidate; its benefit and harm require an ablation.

The ranked memory contains at most 250 videos. Current and previous top-20 candidates are protected, and remaining capacity follows session score with a stable identifier tie-break. Each hint retains at most three temporally deduplicated representative frames per video. Backfilled evidence records the turn at which it was discovered and does not alter earlier committed snapshots.

4 Locked Evaluation

We evaluate 75 owner-approved synthetic progressive queries derived from AIC queries, prior annotations and sampled visual inspection. Approval does not constitute independent continuous-playback verification. A fixed split by connected components of target videos assigns 15 queries (14 groups) to development and 60 queries (55 groups) to evaluation; all three prefixes stay with their query. The corpus contains 1,339,055 InfoShot++ frames per selected visual model across L21–L30.

Code was frozen at commit `b012b9b`. Ground truth, translations, heuristic parser outputs, split, model/index provenance and algorithm configuration were locked before evaluation. Each profile was evaluated once after a complete healthy development run. The same split and parser plans were reused for PE+Qwen. TARA, OCR, speech, audio, expansion and reranking were excluded. Global depth is 200 per model/view. Hint-RRF retains the full observed union before output truncation. Dual-view averages current delta and cumulative evidence without historical memory or rescue. Development pilots at depths 200, 400, 800 and 1000 selected cumulative depth 400 by nearest median latency. Its development p50 remained 0.89s below the PHM reference, so this is not a matched-cost comparison. Non-monotonic pilot latencies also expose live service/cache variation.

The primary metric is video MRR averaged over prefixes within each query and then across queries, with a rank cap of 250 and zero credit for missing targets. Moment Hit@ K evaluates the first preferred frame per displayed video group against half-open accepted intervals, without searching historical frames for an oracle match. Paired uncertainty uses 10,000 bootstrap resamples of target-video

Table 1. PE controlled evaluation on 60 held-out queries. H@1 is video Hit@1 at each hint. MRR is the mean prefix metric; latency is median retrieval latency.

Method	MRR	H1	H@1	H2	H@1	H3	H@1	p50 (ms)
Cumulative	0.6558		0.483		0.583		0.617	504.5
Hint-RRF	0.6093		0.483		0.500		0.483	506.7
Dual-view	0.5980		0.483		0.533		0.500	836.0
PHM without rescue	0.6230		0.483		0.500		0.583	829.9
PHM	0.6459		0.483		0.533		0.600	2131.2
Deeper cumulative	0.6627		0.500		0.600		0.600	532.6

Table 2. PE+Qwen replication on the same held-out queries.

Method	MRR	H1	H@1	H2	H@1	H3	H@1	p50 (ms)
Cumulative	0.6907		0.383		0.633		0.683	1133.6
Hint-RRF	0.6662		0.383		0.517		0.717	1108.6
PHM without rescue	0.6832		0.383		0.567		0.733	1516.9
PHM	0.6903		0.383		0.567		0.733	3875.1

groups, with unadjusted intervals. All 3,330 development/evaluation prefixes were healthy; the held-out tables contain 1,800 prefixes. An artifact audit re-computed metrics from raw snapshots and checked ledger isolation and completeness.

PHM achieves MRR 0.6459 with PE and 0.6903 with PE+Qwen, compared with cumulative retrieval at 0.6558 and 0.6907. Paired differences are -0.0099 (95% interval $[-0.0576, 0.0358]$) and -0.0004 ($[-0.0460, 0.0431]$), respectively. These results do not establish an advantage over cumulative retrieval, nor do they establish equivalence. PHM exceeds Hint-RRF by 0.0367 ($[0.0064, 0.0704]$) with PE and 0.0241 ($[0.0029, 0.0443]$) with PE+Qwen, under the exploratory unadjusted comparisons.

Rescue increases MRR over PHM without rescue by 0.0229 with PE and 0.0071 with PE+Qwen, but both intervals include zero. No target is recovered from outside the rank cap; rescue worsens target rank in 20 PE and 10 PE+Qwen prefixes. PE+Qwen final-turn Hit@1 is 44/60 for PHM versus 41/60 for cumulative; the no-rescue variant also reaches 44/60. This descriptive final-turn difference cannot therefore be attributed to rescue. Final-turn Moment Hit@1 is 0.367 versus 0.400 for PHM/cumulative with PE, and 0.400 for both with PE+Qwen.

The additional retrieval has substantial cost: PHM median latency is 2.13s versus 0.50s for cumulative with PE, and 3.88s versus 1.13s with PE+Qwen. Deeper cumulative retrieval has the highest PE MRR (0.6627). Complete prefix metrics, localization, first/stable-correct turn distributions, rescue counts, work counters, p95 latency, locks and artifact hashes accompany the repository report in [benchmarks/progressive/results/paper-20260921](#).

5 Limitations and Deployment Status

Correctness tests cover hint deduplication and edits, reveal isolation, first-turn ordering, pooling across local/global retrieval, late revisions, failure preservation, provenance and identity-based UI selection. Such tests do not establish effectiveness, usability or competition readiness. A separate PE-only InfoShot++ browser smoke test exercised two hints, eight survivor rescue and four newcomer backfill requests, and verified video seeking to the selected evidence time. This smoke test predates the global-frontier scoring revision; it checks the interaction path, not the revised scoring or an evaluated task. Video-wide evidence may be dispersed across unrelated scenes, and rescue may reinforce a wrong candidate. The operator must still verify an answer moment.

V3C and the other competition collections require separate media registries, temporal mappings, index coverage and submission checks. Their full ingestion, external evaluation, operator study and budgeted inspection are planned work. No calibrated confidence, learned stopping policy or automatic submission is part of PHM v1.

6 Conclusion

The implemented extension makes per-hint candidate evidence explicit and provides bounded in-video retrieval with consistent score pooling and inspectable temporal provenance. Locked experiments show gains over independent-hint rank fusion but no demonstrated mean-prefix advantage over cumulative retrieval, at higher latency. These mixed findings support the implemented interaction mechanism while limiting claims of retrieval superiority. Competition-scale generalization and operator usefulness remain future work.